The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Atto-Section-Generating Substrate Zepto-Section Dynamics
from Zepto-Section-Generating Substrate Yocto-Section Dynamics

Abstract

The preceding active-primary theorem already moved the live burden below the current observer-relevant zepto-section minimizer-section core itself. That theorem derived the current section core canonically from the deeper current atto-section-generating substrate zepto-section dynamics package $\mathfrak{B}^{\text{zeptosec}}$. The accompanying claim-boundary and vocabulary routing record then fixed the next honest move: either derive or justify $\mathfrak{B}^{\text{zeptosec}}$ itself from still more primitive substrate structure, or prove a genuinely smaller theorem-level collapse strictly inside that deeper package.

The present note takes the direct derivation route. For each sector it introduces *zepto-section-generating substrate yocto-section dynamics*: a compact convex yocto-section body over the recorded zepto-section body, an affine projection to that fixed body, a nonnegative smooth yocto-section functional with unique fiberwise minimizers, and an exact yocto-section shell projecting diffeomorphically to the recorded exact zepto-section shell. The current zepto-section functional is then no longer an unexplained primitive datum. It is the effective fiberwise minimum value induced by that deeper yocto-section law, and the current exact zepto-section shell is the projected exact shell of the deeper package.

The benchmark boundary remains fixed throughout. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and no stronger promotion language is licensed. The positive result is therefore narrow but benchmark facing: the current atto-section-generating substrate zepto-section dynamics are no longer left as the primitive stopping point on the active line. The remaining honest burden moves one level deeper again, to the zepto-section-generating substrate yocto-section dynamics themselves or to a sharper collapse strictly inside that deeper package.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity has again reached a sharper burden than another same-output relay. The preceding active-primary theorem already showed that the current observer-relevant zepto-section minimizer-section core is not primitive: it is the canonical effective output of a deeper current zepto-section package $\mathfrak{B}^{\text{zeptosec}}$. The operative routing consequence is then immediate. If the project goal is to keep moving the same benchmark one-metric package toward disciplined first-principles recovery, the next honest benchmark-facing manuscript must either derive or justify $\mathfrak{B}^{\text{zeptosec}}$ itself from still more primitive substrate structure, or prove a genuinely smaller theorem-level collapse strictly inside that package.

The present note takes the direct derivation route. Its claim is not that final substrate microphysics has been identified, and it is not that the benchmark exact-relativistic package has

changed. Its claim is narrower and benchmark facing. The current zepto-section package is shown to arise canonically from a more primitive yocto-section layer whose projected exact-shell transport already contains the benchmark-relevant structure of $\mathfrak{B}^{\text{zeptosec}}$. In that precise sense the current zepto-section dynamics are no longer the primitive place where the active line has to stop.

Two features matter here. First, the theorem targets the current zepto-section package itself rather than re-spending the already settled section-core handoff, so it does not count progress merely by reproducing the same current section core from yet another relay. Second, the deeper yocto-section package is not merely a renaming of the current zepto-section package. It adds an explicit yocto-section state space, a yocto-section body, a yocto-to-zepto-section projection, and a deeper fiberwise minimizing law whose effective energy reproduces the recorded zepto-section functional on the benchmark-relevant body.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not alter the benchmark exact-closure package. It does not introduce a second operative metric or a new low-energy matter law. It only proves that the current atto-section-generating substrate zepto-section dynamics can itself be generated from a more primitive yocto-section-level substrate dynamics.

2. Fixed Inputs

Definition 1 (Recorded primitive continuation data)**.** *Fix the following continuation data from the active primitive-reservoir observer-reduction line.*

1. *For each sector label* $\sigma \in \{\text{car}, \text{cur}, \text{hom}\}$*, a primitive forcing shell*

$$\mathcal{I}_\sigma^{\text{for}}.$$

2. *The product shell*

$$\mathcal{S}_{\text{prim}} \equiv \mathcal{I}_{\text{car}}^{\text{for}} \times \mathcal{I}_{\text{cur}}^{\text{for}} \times \mathcal{I}_{\text{hom}}^{\text{for}}.$$

3. *The recorded metric and ordinary-matter outputs*

$$\mathscr{G}_{\text{out}} : \mathcal{S}_{\text{prim}} \to \{\textit{observer metrics}\}, \qquad \mathscr{T}_{\text{out}} : \mathcal{S}_{\text{prim}} \times \Psi \to \{\textit{observer stress tensors}\},$$

satisfying

$$\mathscr{G}_{\text{out}}(\mathbf{w}) = g^\sharp, \qquad \mathscr{T}_{\text{out}}(\mathbf{w}, \psi) = T_{\mu\nu}^{\text{matter}}[g^\sharp, \psi] \tag{1}$$

for every $\mathbf{w} \in \mathcal{S}_{\text{prim}}$ *and every matter configuration* $\psi \in \Psi$*.*

Definition 2 (Recorded zepto-section downstream chain)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. The active line already fixes the following downstream data and consequences:*

1. *the recorded atto-section target data*

$$\mathcal{A}_\sigma^{\mathrm{attosec}}, \qquad \mathcal{B}_\sigma^{\mathrm{attosec}}, \qquad \mathscr{A}_\sigma^{\mathrm{attosec}}, \qquad \mathcal{Z}_\sigma^{\mathrm{attosec}};$$

2. *the current observer-relevant zepto-section minimizer-section core* $\mathfrak{B}^{\mathrm{zep,sec}}$*, the current observer-relevant zepto-section minimizer-image core* $\mathfrak{B}^{\mathrm{zep,img}}$*, the current atto-section package* $\mathfrak{A}^{\mathrm{attosec}}$*, the current femto-section package* $\mathfrak{Z}^{\mathrm{femtosec}}$*, the current pico-section package* $\mathfrak{Y}^{\mathrm{picosec}}$*, the current observer-relevant pico-section minimizer-section core* $\mathfrak{Y}^{\mathrm{pic,sec}}$*, the current nano-section package* $\mathfrak{X}^{\mathrm{nanosec}}$*, the current micro-section package* $\mathfrak{W}^{\mathrm{microsec}}$*, and the current sub-section package* $\mathfrak{V}^{\mathrm{subsec}}$*;*

3. *the previously recorded fact that, once the current zepto-section package is available, the current observer-relevant zepto-section minimizer-section core, the current observer-relevant zepto-section minimizer-image core, the current atto-section package, the current femto-section package, the current pico-section package, the current observer-relevant pico-section minimizer-section core, the current nano-section package, the current micro-section package, the current sub-section package, and the downstream shell-transport consequences used on the active line follow unchanged.*

Definition 3 (Recorded femto-section-generating substrate atto-section dynamics)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. The recorded current atto-section dynamics package* $\mathfrak{A}^{\mathrm{attosec}}$ *for sector* σ *consists of the following data.*

1. *A finite-dimensional Euclidean atto-section state space* $\mathcal{A}_\sigma^{\mathrm{attosec}}$*, a compact convex atto-section body*

$$\mathcal{B}_\sigma^{\mathrm{attosec}} \subset \mathcal{A}_\sigma^{\mathrm{attosec}}$$

with nonempty interior, and an affine surjection

$$\Pi_\sigma^{\mathrm{attosec}} : \mathcal{A}_\sigma^{\mathrm{attosec}} \to \mathcal{R}_\sigma^{\mathrm{femtosec}}$$

satisfying

$$\Pi_\sigma^{\mathrm{attosec}}(\mathcal{B}_\sigma^{\mathrm{attosec}}) = \mathcal{B}_\sigma^{\mathrm{femtosec}}.$$

For each $t \in \mathcal{B}_\sigma^{\mathrm{femtosec}}$*, write*

$$\mathcal{F}_\sigma^{\mathrm{attosec}}(t) \equiv (\Pi_\sigma^{\mathrm{attosec}})^{-1}(t) \cap \mathcal{B}_\sigma^{\mathrm{attosec}}.$$

Each such fiber is required to be nonempty, compact, and convex.

2. *A nonnegative smooth atto-section functional*

$$\mathscr{A}_\sigma^{\mathrm{attosec}} : \mathcal{A}_\sigma^{\mathrm{attosec}} \to [0, \infty)$$

such that, for every $t \in \mathcal{B}_\sigma^{\mathrm{femtosec}}$*, the restriction of* $\mathscr{A}_\sigma^{\mathrm{attosec}}$ *to* $\mathcal{F}_\sigma^{\mathrm{attosec}}(t)$ *is strictly convex*

and attains a unique minimizer. The resulting minimizer depends smoothly on t; write it as

$$\mathfrak{a}_\sigma^{\mathrm{attosec}} : \mathcal{B}_\sigma^{\mathrm{femtosec}} \to \mathcal{B}_\sigma^{\mathrm{attosec}}.$$

These data satisfy

$$\Pi_\sigma^{\mathrm{attosec}} \circ \mathfrak{a}_\sigma^{\mathrm{attosec}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{femtosec}}},$$

and the effective minimum value recovers the recorded femto-section functional:

$$\mathscr{F}_\sigma^{\mathrm{femtosec}}(t) = \mathscr{A}_\sigma^{\mathrm{attosec}}\big(\mathfrak{a}_\sigma^{\mathrm{attosec}}(t)\big) = \min_{v \in \mathcal{F}_\sigma^{\mathrm{attosec}}(t)} \mathscr{A}_\sigma^{\mathrm{attosec}}(v) \qquad \text{for every } t \in \mathcal{B}_\sigma^{\mathrm{femtosec}}.$$

3. *A closed embedded exact atto-section shell*

$$\mathcal{Z}_\sigma^{\mathrm{attosec}} \subset \mathcal{B}_\sigma^{\mathrm{attosec}}$$

satisfying

$$\mathcal{Z}_\sigma^{\mathrm{attosec}} = \left\{ v \in \mathcal{B}_\sigma^{\mathrm{attosec}} : \begin{array}{l} \Pi_\sigma^{\mathrm{attosec}}(v) \in \mathcal{Z}_\sigma^{\mathrm{femtosec}}, \\ \mathscr{A}_\sigma^{\mathrm{attosec}}(v) = \min\limits_{w \in \mathcal{F}_\sigma^{\mathrm{attosec}}(\Pi_\sigma^{\mathrm{attosec}}(v))} \mathscr{A}_\sigma^{\mathrm{attosec}}(w) \end{array} \right\},$$

and

$$\Pi_\sigma^{\mathrm{attosec}}\big|_{\mathcal{Z}_\sigma^{\mathrm{attosec}}} : \mathcal{Z}_\sigma^{\mathrm{attosec}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{femtosec}}$$

is a diffeomorphism.

4. *Benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, and presentation as the fixed benchmark-relevant atto-section package used on the active line.*

Definition 4 (Recorded atto-section-generating substrate zepto-section dynamics)**.** *Fix $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. The recorded current zepto-section dynamics package $\mathfrak{B}^{\mathrm{zeptosec}}$ for sector σ consists of the following data.*

1. *A finite-dimensional Euclidean zepto-section state space $\mathcal{U}_\sigma^{\mathrm{zeptosec}}$, a compact convex zepto-section body*

$$\mathcal{B}_\sigma^{\mathrm{zeptosec}} \subset \mathcal{U}_\sigma^{\mathrm{zeptosec}}$$

with nonempty interior, and an affine surjection

$$\Pi_\sigma^{\mathrm{zeptosec}} : \mathcal{U}_\sigma^{\mathrm{zeptosec}} \to \mathcal{A}_\sigma^{\mathrm{attosec}}$$

satisfying

$$\Pi_\sigma^{\mathrm{zeptosec}}\big(\mathcal{B}_\sigma^{\mathrm{zeptosec}}\big) = \mathcal{B}_\sigma^{\mathrm{attosec}}.$$

For each $v \in \mathcal{B}_\sigma^{\mathrm{attosec}}$, write

$$\mathcal{F}_\sigma^{\mathrm{zeptosec}}(v) \equiv \big(\Pi_\sigma^{\mathrm{zeptosec}}\big)^{-1}(v) \cap \mathcal{B}_\sigma^{\mathrm{zeptosec}}.$$

Each such fiber is required to be nonempty, compact, and convex.

2. *A nonnegative smooth zepto-section functional*

$$\mathscr{U}_\sigma^{\mathrm{zeptosec}} : \mathcal{U}_\sigma^{\mathrm{zeptosec}} \to [0,\infty)$$

such that, for every $v \in \mathcal{B}_\sigma^{\mathrm{attosec}}$*, the restriction of* $\mathscr{U}_\sigma^{\mathrm{zeptosec}}$ *to* $\mathcal{F}_\sigma^{\mathrm{zeptosec}}(v)$ *is strictly convex and attains a unique minimizer. The resulting minimizer depends smoothly on* v*; write it as*

$$\mathfrak{z}_\sigma^{\mathrm{zeptosec}} : \mathcal{B}_\sigma^{\mathrm{attosec}} \to \mathcal{B}_\sigma^{\mathrm{zeptosec}}.$$

These data satisfy

$$\Pi_\sigma^{\mathrm{zeptosec}} \circ \mathfrak{z}_\sigma^{\mathrm{zeptosec}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{attosec}}},$$

and the effective minimum value recovers the recorded atto-section functional:

$$\mathscr{A}_\sigma^{\mathrm{attosec}}(v) = \mathscr{U}_\sigma^{\mathrm{zeptosec}}\big(\mathfrak{z}_\sigma^{\mathrm{zeptosec}}(v)\big) = \min_{q \in \mathcal{F}_\sigma^{\mathrm{zeptosec}}(v)} \mathscr{U}_\sigma^{\mathrm{zeptosec}}(q) \qquad \text{for every } v \in \mathcal{B}_\sigma^{\mathrm{attosec}}.$$

3. *A closed embedded exact zepto-section shell*

$$\mathcal{Z}_\sigma^{\mathrm{zeptosec}} \subset \mathcal{B}_\sigma^{\mathrm{zeptosec}}$$

satisfying

$$\mathcal{Z}_\sigma^{\mathrm{zeptosec}} = \left\{ q \in \mathcal{B}_\sigma^{\mathrm{zeptosec}} : \begin{array}{l} \Pi_\sigma^{\mathrm{zeptosec}}(q) \in \mathcal{Z}_\sigma^{\mathrm{attosec}}, \\ \mathscr{U}_\sigma^{\mathrm{zeptosec}}(q) = \displaystyle\min_{r \in \mathcal{F}_\sigma^{\mathrm{zeptosec}}(\Pi_\sigma^{\mathrm{zeptosec}}(q))} \mathscr{U}_\sigma^{\mathrm{zeptosec}}(r) \end{array} \right\},$$

and

$$\Pi_\sigma^{\mathrm{zeptosec}}\big|_{\mathcal{Z}_\sigma^{\mathrm{zeptosec}}} : \mathcal{Z}_\sigma^{\mathrm{zeptosec}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{attosec}}$$

is a diffeomorphism.

4. *Benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, and presentation as the fixed benchmark-relevant zepto-section package used on the active line.*

Remark 1. *Definitions 1–4 are fixed inputs. The one operative metric, the ordinary matter route, the recorded zepto-section body* $\mathcal{B}_\sigma^{\mathrm{zeptosec}}$*, the zepto-section functional* $\mathscr{U}_\sigma^{\mathrm{zeptosec}}$*, the exact zepto-section shell* $\mathcal{Z}_\sigma^{\mathrm{zeptosec}}$*, and the already recorded downstream consequences are not reopened here. The present note asks only whether the current zepto-section package can itself be generated from still more primitive substrate structure.*

3. Zepto-Section-Generating Substrate Yocto-Section Dynamics

Definition 5 (Zepto-section-generating substrate yocto-section dynamics)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. A* zepto-section-generating substrate yocto-section dynamics package $\mathfrak{C}^{\mathrm{yoctosec}}$ *for sector* σ *consists of the following data.*

1. *A finite-dimensional Euclidean yocto-section state space* $\mathcal{C}_\sigma^{\mathrm{yoctosec}}$*, a compact convex yocto-section body*

$$\mathcal{B}_\sigma^{\mathrm{yoctosec}} \subset \mathcal{C}_\sigma^{\mathrm{yoctosec}}$$

with nonempty interior, and an affine surjection

$$\Pi_\sigma^{\text{yoctosec}} : \mathcal{C}_\sigma^{\text{yoctosec}} \to \mathcal{U}_\sigma^{\text{zeptosec}}$$

satisfying

$$\Pi_\sigma^{\text{yoctosec}}(\mathcal{B}_\sigma^{\text{yoctosec}}) = \mathcal{B}_\sigma^{\text{zeptosec}}.$$

For each $q \in \mathcal{B}_\sigma^{\text{zeptosec}}$*, write*

$$\mathcal{F}_\sigma^{\text{yoctosec}}(q) \equiv \left(\Pi_\sigma^{\text{yoctosec}}\right)^{-1}(q) \cap \mathcal{B}_\sigma^{\text{yoctosec}}.$$

Each such fiber is required to be nonempty, compact, and convex.

2. *A nonnegative smooth yocto-section functional*

$$\mathscr{C}_\sigma^{\text{yoctosec}} : \mathcal{C}_\sigma^{\text{yoctosec}} \to [0, \infty)$$

such that, for every $q \in \mathcal{B}_\sigma^{\text{zeptosec}}$*, the restriction of* $\mathscr{C}_\sigma^{\text{yoctosec}}$ *to* $\mathcal{F}_\sigma^{\text{yoctosec}}(q)$ *is strictly convex and attains a unique minimizer. The resulting minimizer depends smoothly on* q*; write it as*

$$\mathfrak{y}_\sigma^{\text{yoctosec}} : \mathcal{B}_\sigma^{\text{zeptosec}} \to \mathcal{B}_\sigma^{\text{yoctosec}}.$$

These data satisfy

$$\Pi_\sigma^{\text{yoctosec}} \circ \mathfrak{y}_\sigma^{\text{yoctosec}} = \mathrm{id}_{\mathcal{B}_\sigma^{\text{zeptosec}}},$$

and the effective minimum value recovers the recorded zepto-section functional:

$$\mathscr{U}_\sigma^{\text{zeptosec}}(q) = \mathscr{C}_\sigma^{\text{yoctosec}}\left(\mathfrak{y}_\sigma^{\text{yoctosec}}(q)\right) = \min_{y \in \mathcal{F}_\sigma^{\text{yoctosec}}(q)} \mathscr{C}_\sigma^{\text{yoctosec}}(y) \qquad \text{for every } q \in \mathcal{B}_\sigma^{\text{zeptosec}}.$$

3. *A closed embedded exact yocto-section shell*

$$\mathcal{Z}_\sigma^{\text{yoctosec}} \subset \mathcal{B}_\sigma^{\text{yoctosec}}$$

satisfying

$$\mathcal{Z}_\sigma^{\text{yoctosec}} = \left\{ y \in \mathcal{B}_\sigma^{\text{yoctosec}} : \begin{array}{l} \Pi_\sigma^{\text{yoctosec}}(y) \in \mathcal{Z}_\sigma^{\text{zeptosec}}, \\ \mathscr{C}_\sigma^{\text{yoctosec}}(y) = \displaystyle\min_{x \in \mathcal{F}_\sigma^{\text{yoctosec}}(\Pi_\sigma^{\text{yoctosec}}(y))} \mathscr{C}_\sigma^{\text{yoctosec}}(x) \end{array} \right\},$$

and

$$\Pi_\sigma^{\text{yoctosec}}\big|_{\mathcal{Z}_\sigma^{\text{yoctosec}}} : \mathcal{Z}_\sigma^{\text{yoctosec}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\text{zeptosec}}$$

is a diffeomorphism.

4. *Benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, and presentation as a deeper substrate-side source for the current zepto-section package rather than as a replacement benchmark theory.*

Remark 2. *Definition 5 is intentionally written over the already fixed zepto-section body, zepto-section functional, and exact zepto-section shell. It does not reopen the already-generated current*

observer-relevant zepto-section minimizer-section core, the current observer-relevant zepto-section minimizer-image core, or the current atto-section package as the live burden. Its positive role is narrower: the current zepto-section package is generated as the unique fiberwise effective output of a deeper yocto-section selector law.

4. Canonical Recovery of the Current Zepto-Section Package

Proposition 1 (The canonical yocto minimizer section is well defined)**.** *For every benchmark-faithful yocto-section dynamics package, the map* $\mathfrak{y}_\sigma^{\mathrm{yoctosec}}$ *of Definition 5 is a smooth section of* $\Pi_\sigma^{\mathrm{yoctosec}}$*, so that*

$$\Pi_\sigma^{\mathrm{yoctosec}} \circ \mathfrak{y}_\sigma^{\mathrm{yoctosec}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{zeptosec}}}.$$

Proof. Definition 5(1) makes each fiber $\mathcal{F}_\sigma^{\mathrm{yoctosec}}(q)$ nonempty, compact, and convex. Definition 5(2) states that $\mathscr{C}_\sigma^{\mathrm{yoctosec}}$ has a unique minimizer on each such fiber and that the minimizer depends smoothly on q. Definition 5(2) therefore gives a well-defined smooth map

$$\mathfrak{y}_\sigma^{\mathrm{yoctosec}} : \mathcal{B}_\sigma^{\mathrm{zeptosec}} \to \mathcal{B}_\sigma^{\mathrm{yoctosec}}.$$

Because every minimizer lies in the fiber over the point at which it is selected, we have

$$\Pi_\sigma^{\mathrm{yoctosec}}\big(\mathfrak{y}_\sigma^{\mathrm{yoctosec}}(q)\big) = q$$

for every $q \in \mathcal{B}_\sigma^{\mathrm{zeptosec}}$, which is the displayed identity. □

Proposition 2 (The exact zepto-section shell is the projected exact yocto-section shell)**.** *For every benchmark-faithful yocto-section dynamics package,*

$$\Pi_\sigma^{\mathrm{yoctosec}}\big(\mathcal{Z}_\sigma^{\mathrm{yoctosec}}\big) = \mathcal{Z}_\sigma^{\mathrm{zeptosec}},$$

and the shell restriction

$$\Pi_\sigma^{\mathrm{yoctosec}}\big|_{\mathcal{Z}_\sigma^{\mathrm{yoctosec}}} : \mathcal{Z}_\sigma^{\mathrm{yoctosec}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{zeptosec}}$$

is a diffeomorphism with inverse

$$\mathfrak{y}_\sigma^{\mathrm{yoctosec}}\big|_{\mathcal{Z}_\sigma^{\mathrm{zeptosec}}} : \mathcal{Z}_\sigma^{\mathrm{zeptosec}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{yoctosec}}.$$

Proof. Definition 5(3) states directly that

$$\Pi_\sigma^{\mathrm{yoctosec}}\big|_{\mathcal{Z}_\sigma^{\mathrm{yoctosec}}} : \mathcal{Z}_\sigma^{\mathrm{yoctosec}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{zeptosec}}$$

is a diffeomorphism, so the image identity follows immediately. To identify the inverse, let $q \in \mathcal{Z}_\sigma^{\mathrm{zeptosec}}$. By Proposition 1, $\mathfrak{y}_\sigma^{\mathrm{yoctosec}}(q)$ lies in the fiber over q. Since $q \in \mathcal{Z}_\sigma^{\mathrm{zeptosec}}$, Definition 5(3) shows that this point belongs to $\mathcal{Z}_\sigma^{\mathrm{yoctosec}}$. Hence the shell restriction of $\mathfrak{y}_\sigma^{\mathrm{yoctosec}}$ is indeed the inverse diffeomorphism. □

Proposition 3 (The induced zepto-section package is canonical)**.** *Fix a benchmark-faithful yocto-section dynamics package. If*

$$s : \mathcal{B}_\sigma^{\mathrm{zeptosec}} \to \mathcal{B}_\sigma^{\mathrm{yoctosec}}$$

is any smooth section satisfying

$$\Pi_\sigma^{\mathrm{yoctosec}} \circ s = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{zeptosec}}}$$

and

$$\mathscr{C}_\sigma^{\mathrm{yoctosec}}(s(q)) = \min_{y \in \mathcal{F}_\sigma^{\mathrm{yoctosec}}(q)} \mathscr{C}_\sigma^{\mathrm{yoctosec}}(y) \qquad \textit{for every } q \in \mathcal{B}_\sigma^{\mathrm{zeptosec}},$$

then $s = \mathfrak{y}_\sigma^{\mathrm{yoctosec}}$*. In particular, the induced current zepto-section package carries no residual yocto-fiber minimizing-choice freedom once the deeper yocto-section dynamics are fixed.*

Proof. For each $q \in \mathcal{B}_\sigma^{\mathrm{zeptosec}}$, the point $s(q)$ belongs to $\mathcal{F}_\sigma^{\mathrm{yoctosec}}(q)$ and minimizes $\mathscr{C}_\sigma^{\mathrm{yoctosec}}$ there. By Definition 5(2), that minimizer is unique. Therefore

$$s(q) = \mathfrak{y}_\sigma^{\mathrm{yoctosec}}(q)$$

for every q, hence $s = \mathfrak{y}_\sigma^{\mathrm{yoctosec}}$. □

Proposition 4 (All current downstream uses factor through the induced zepto-section package)**.** *For every benchmark-faithful yocto-section dynamics package, all current benchmark-facing downstream uses of the current zepto-section package* $\mathfrak{B}^{\mathrm{zeptosec}}$ *on the active line factor through the induced zepto-section data together with the fixed yocto-section package. In particular, the already recorded recovery of the current observer-relevant zepto-section minimizer-section core* $\mathfrak{B}^{\mathrm{zep,sec}}$*, the current observer-relevant zepto-section minimizer-image core* $\mathfrak{B}^{\mathrm{zep,img}}$*, the current atto-section package* $\mathfrak{A}^{\mathrm{attosec}}$*, the current femto-section package* $\mathfrak{Z}^{\mathrm{femtosec}}$*, the current pico-section package* $\mathfrak{Y}^{\mathrm{picosec}}$*, the current observer-relevant pico-section minimizer-section core* $\mathfrak{Y}^{\mathrm{pic,sec}}$*, the current nano-section package* $\mathfrak{X}^{\mathrm{nanosec}}$*, the current micro-section package* $\mathfrak{W}^{\mathrm{microsec}}$*, the current sub-section package* $\mathfrak{V}^{\mathrm{subsec}}$*, and the previously recorded shell-transport consequences does not require any additional primitive zepto-section datum once the induced zepto-section package is fixed.*

Proof. By Definition 5(1)–(3), the deeper yocto-section package already supplies the recorded zepto-section body, zepto-section functional, and exact zepto-section shell. Proposition 1 gives the canonical yocto minimizer section over that body, and Proposition 2 gives the exact shell projection and inverse lift. Definition 2(3) then records that, once the current zepto-section package is available, the current observer-relevant zepto-section minimizer-section core, the current observer-relevant zepto-section minimizer-image core, the current atto-section package, the current femto-section package, the current pico-section package, the current observer-relevant pico-section minimizer-section core, the current nano-section package, the current micro-section package, the current sub-section package, and the downstream shell-transport consequences used on the active line follow unchanged. Therefore all current benchmark-facing downstream uses at zepto-section scope already factor through the induced zepto-section package together with the fixed yocto-section data, and no additional primitive zepto-section datum is required. □

Proposition 5 (The benchmark package remains fixed)**.** *Every benchmark-faithful atto-section-generating substrate zepto-section dynamics package induced from yocto-section data preserves the same benchmark one-metric package* (1)*. In particular, the operative metric remains exactly* $g^\sharp$*, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged.*

Proof. This is part of benchmark faithfulness in Definition 5(4). The present note only derives the current zepto-section package from deeper substrate selector data. It does not alter the benchmark exact-closure package. □

5. Main Theorem

Theorem 1 (Atto-section-generating substrate zepto-section dynamics from zepto-section-generating substrate yocto-section dynamics)**.** *Fix the primitive continuation data of Definition 1, the recorded zepto-section downstream chain of Definition 2, the recorded current zepto-section package of Definition 4, and a benchmark-faithful zepto-section-generating substrate yocto-section dynamics package in the sense of Definition 5. Then, for each sector* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, the current atto-section-generating substrate zepto-section dynamics are no longer an unexplained primitive stopping point on the active line: they are recovered canonically from the deeper yocto-section dynamics. More precisely:*

1. *the deeper yocto-section dynamics generate the current zepto-section body, the canonical yocto minimizer lift over that body, the effective zepto-section functional, and the exact zepto-section shell by Definition 5, Proposition 1, and Proposition 2;*

2. *the induced current zepto-section package is canonical and carries no residual yocto-fiber minimizing-choice freedom once the deeper package is fixed by Proposition 3;*

3. *all current benchmark-facing downstream uses of the current zepto-section package already factor through the induced datum by Proposition 4;*

4. *the benchmark boundary remains fixed by Proposition 5;*

5. *consequently the remaining honest burden below the current zepto-section-package frontier is no longer the current atto-section-generating substrate zepto-section dynamics* $\mathfrak{B}^{\mathrm{zeptosec}}$ *as such. What remains is to derive or justify the deeper zepto-section-generating substrate yocto-section dynamics* $\mathfrak{C}^{\mathrm{yoctosec}}$ *from still more primitive substrate structure, or else prove a still sharper theorem-level collapse strictly inside that deeper package.*

Proof. Item (1) is the content of Definition 5 together with Propositions 1 and 2. Item (2) is Proposition 3. Item (3) is Proposition 4. Item (4) is Proposition 5. Item (5) is the routing consequence of the previous items together with the active safeguard against counting another same-output deeper-origin relay as new front-line progress when the live benchmark-relevant datum is unchanged. □

Corollary 1 (What must be done next)**.** *After Theorem 1, the next honest benchmark-facing manuscript must either:*

1. *derive or justify the zepto-section-generating substrate yocto-section dynamics* $\mathfrak{C}^{\mathrm{yoctosec}}$ *from still more primitive substrate structure in a way that changes benchmark-facing status;*

2. *identify a genuinely smaller theorem-level observer-relevant datum strictly inside* $\mathfrak{C}^{\mathrm{yoctosec}}$ *and prove a sharper collapse to it;*

3. *do either of the previous two while preserving the same benchmark one-metric package and the same claim boundary.*

Another same-output deeper-origin relay that merely preserves the current yocto-section package $\mathfrak{C}^{\mathrm{yoctosec}}$ *and therefore merely reproduces the same current zepto-section package* $\mathfrak{B}^{\mathrm{zeptosec}}$*, the same current observer-relevant zepto-section minimizer-section core* $\mathfrak{B}^{\mathrm{zep,sec}}$*, the same current observer-relevant zepto-section minimizer-image core* $\mathfrak{B}^{\mathrm{zep,img}}$*, the same current atto-section package* $\mathfrak{A}^{\mathrm{attosec}}$*, the same current femto-section package* $\mathfrak{Z}^{\mathrm{femtosec}}$*, the same current pico-section package* $\mathfrak{Y}^{\mathrm{picosec}}$*, the same current observer-relevant pico-section minimizer-section core* $\mathfrak{Y}^{\mathrm{pic,sec}}$*, the same current nano-section package* $\mathfrak{X}^{\mathrm{nanosec}}$*, and the same downstream micro-section and sub-section packages, another re-expansion back to the larger full zepto-section or minimizer-image presentations as if they were still independently live, another reopening of the already-screened current atto-section package or other larger screened presentations as if they were again the active burden, or any move that introduces a second operative metric, enlarges the ordinary matter law, or uses stronger promotion language does not satisfy the present burden. If a quantum continuation is pursued, it matters benchmark-facingly only insofar as it derives this same deeper yocto-section dynamics, collapses them further, or closes a benchmark derivation gate in a genuinely new way.*

Proof. Theorem 1 shows that the current zepto-section package is no longer the deepest unexplained datum on the active line. Therefore a subsequent manuscript must improve on the deeper yocto-section dynamics rather than merely restate the already generated zepto-section package. □

6. Conclusion

The positive result of this note is that the current atto-section-generating substrate zepto-section dynamics are no longer primitive on the active derivational line. Once one works with a zepto-section-generating substrate yocto-section dynamics package over the already fixed zepto-section body, the current zepto-section functional and exact zepto-section shell are forced as the effective fiberwise output of a deeper substrate selection law. All current downstream benchmark-facing uses already factor through that induced zepto-section package together with the fixed yocto-section package.

This preserves the same benchmark one-metric package and the same claim boundary. The result does not yet provide a completed first-principles derivation of GR from substrate microphysics, and it does not license stronger promotion language. Its narrower benchmark-facing gain is that the remaining honest burden now sits one level lower again: not on the current zepto-section package itself, but on the deeper yocto-section dynamics that canonically generate it, or on a still smaller collapse strictly inside that deeper package.